

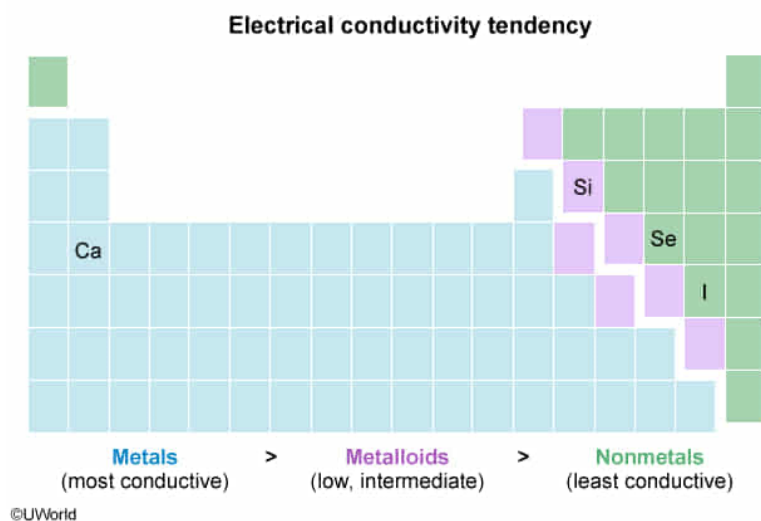
Considering the type of element represented by each of the following, which of the elements below is most likely to have the highest electrical conductivity?

- ✓ ☒ A. Ca [63%]  
☐ B. Si [17%]  
☐ C. Se [6%]  
☐ D. I [12%]

Correct

63% answered correctly

### Explanation:



Within the organization of the [periodic table](#), the elements can be broadly classified as metals, metalloids, and nonmetals. The heavy zigzag line shown on some periodic tables separates metals from nonmetals, with **metals** located to the left of the line and **nonmetals** to the right. The [metalloids](#) are located along the line (excluding Al and Po).

Although these divisions are imperfect because they broadly group many different elements with individual properties, they are useful in categorizing elements according to many shared

characteristics. Metals tend to be shiny, ductile, and good conductors of heat and electricity whereas nonmetals tend to be dull, brittle, and poor electrical conductors (making them good electrical insulators). Metalloids share some characteristics of metals and some characteristics of nonmetals. As a result, metalloids tend to be better electrical conductors than nonmetals but not as conductive as metals. This makes metalloids useful as semiconductors in some instances and as insulators in others.

No consistent trend for electrical conductivity holds true across the periodic table within periods (rows) or groups (columns). However, on the basis of elemental types, calcium (a metal) can be expected to be more conductive than silicon (a metalloid). Likewise, calcium can be expected to be much more conductive than selenium and iodine (both nonmetals), as [verified by property comparisons](#). Although exceptions do exist when comparing two particular elements using this generalized approach, these broad categories of elements are helpful in **qualitative comparisons**.

**(Choice B)** Silicon is a metalloid. Metalloids tend to be more conductive than nonmetals but less conductive than metals.

**(Choices C and D)** Selenium and iodine are both nonmetals. Nonmetals tend to be the least conductive when compared to metalloids and metals.

### **Educational objective:**

Elements can be broadly classified on the periodic table as metals, metalloids, or nonmetals. Due to many shared characteristics within each of these categories, the general traits of each type of element are often useful for qualitative comparisons between elements.